

NumericalAlgebra 4

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PROBLEM I Use the Taylor series to show that the approximation

$$\frac{\partial}{\partial x} \left(a \frac{\partial u}{\partial x} \right) (x_i, y_j) \approx \frac{a_{i+\frac{1}{2},j}(u_{i+1,j} - u_{i,j}) - a_{i-\frac{1}{2},j}(u_{i,j} - u_{i-1,j})}{h_x^2}$$

is second-order accurate, provided that

$$a \frac{\partial u}{\partial x} \in C^3, \quad a \frac{\partial^3 u}{\partial x^3} \in C^1.$$

That is, show that the absolute value of the difference between the right-hand side and left-hand side is bounded by

$$\frac{h_x^2}{24} \left[\max_{x \in [x_{i-1}, x_{i+1}]} \left| \frac{\partial^3}{\partial x^3} \left(a \frac{\partial u}{\partial x} \right) \right| + \max_{x \in [x_{i-1}, x_{i+1}]} \left| \frac{\partial}{\partial x} \left(a \frac{\partial^3 u}{\partial x^3} \right) \right| \right].$$

Lemma 1 (Midpoint Taylor expansion). Let u be sufficiently smooth on $[x_i, x_{i+1}]$, and let

$$x_{i+\frac{1}{2}} = x_i + \frac{h}{2}, x_{i+1} = x_i + h.$$

Then

$$\frac{u_{i+1,j} - u_{i,j}}{h} = u_x(x_{i+\frac{1}{2}}, y_j) + \frac{h^2}{24} u_{xxx}(x_{i+\frac{1}{2}}, y_j) + O(h^4).$$

Similarly,

$$\frac{u_{i,j} - u_{i-1,j}}{h} = u_x(x_{i-\frac{1}{2}}, y_j) + \frac{h^2}{24} u_{xxx}(x_{i-\frac{1}{2}}, y_j) + O(h^4).$$

证明. We prove the first formula. Let

$$m = x_{i+\frac{1}{2}}.$$

Then

$$x_{i+1} = m + \frac{h}{2}, \quad x_i = m - \frac{h}{2}.$$

Taylor expansion about m gives

$$u(x_{i+1}, y_j) = u(m, y_j) + \frac{h}{2} u_x(m, y_j) + \frac{h^2}{8} u_{xx}(m, y_j) + \frac{h^3}{48} u_{xxx}(m, y_j) + O(h^4),$$

and

$$u(x_i, y_j) = u(m, y_j) - \frac{h}{2}u_x(m, y_j) + \frac{h^2}{8}u_{xx}(m, y_j) - \frac{h^3}{48}u_{xxx}(m, y_j) + O(h^4).$$

Subtracting the two expansions yields

$$u_{i+1,j} - u_{i,j} = hu_x(m, y_j) + \frac{h^3}{24}u_{xxx}(m, y_j) + O(h^5).$$

Dividing by h , we obtain

$$\frac{u_{i+1,j} - u_{i,j}}{h} = u_x(x_{i+\frac{1}{2}}, y_j) + \frac{h^2}{24}u_{xxx}(x_{i+\frac{1}{2}}, y_j) + O(h^4).$$

The second formula is proved in the same way. □

SECTION . [Proof of the approximation] Fix $y = y_j$, and write

$$h = h_x.$$

Define

$$F(x) = a(x, y_j)u_x(x, y_j), \quad G(x) = a(x, y_j)u_{xxx}(x, y_j).$$

Then

$$F'(x_i) = \frac{\partial}{\partial x} \left(a \frac{\partial u}{\partial x} \right) (x_i, y_j).$$

By the midpoint Taylor expansion lemma,

$$\frac{u_{i+1,j} - u_{i,j}}{h} = u_x(x_{i+\frac{1}{2}}, y_j) + \frac{h^2}{24}u_{xxx}(x_{i+\frac{1}{2}}, y_j) + O(h^4),$$

and

$$\frac{u_{i,j} - u_{i-1,j}}{h} = u_x(x_{i-\frac{1}{2}}, y_j) + \frac{h^2}{24}u_{xxx}(x_{i-\frac{1}{2}}, y_j) + O(h^4).$$

Multiplying by $a_{i+\frac{1}{2},j}$ and $a_{i-\frac{1}{2},j}$, respectively, gives

$$a_{i+\frac{1}{2},j} \frac{u_{i+1,j} - u_{i,j}}{h} = F(x_{i+\frac{1}{2}}) + \frac{h^2}{24}G(x_{i+\frac{1}{2}}) + O(h^4),$$

and

$$a_{i-\frac{1}{2},j} \frac{u_{i,j} - u_{i-1,j}}{h} = F(x_{i-\frac{1}{2}}) + \frac{h^2}{24}G(x_{i-\frac{1}{2}}) + O(h^4).$$

Subtracting and dividing by h , we obtain

$$\begin{aligned} & \frac{a_{i+\frac{1}{2},j}(u_{i+1,j} - u_{i,j}) - a_{i-\frac{1}{2},j}(u_{i,j} - u_{i-1,j})}{h^2} \\ &= \frac{F(x_{i+\frac{1}{2}}) - F(x_{i-\frac{1}{2}})}{h} + \frac{h^2}{24} \frac{G(x_{i+\frac{1}{2}}) - G(x_{i-\frac{1}{2}})}{h} + O(h^3). \end{aligned}$$

Now, by the centered difference formula,

$$\frac{F(x_{i+\frac{1}{2}}) - F(x_{i-\frac{1}{2}})}{h} = F'(x_i) + \frac{h^2}{24}F'''(\eta)$$

for some

$$\eta \in (x_{i-1}, x_{i+1}).$$

Also, by the mean value theorem,

$$\frac{G(x_{i+\frac{1}{2}}) - G(x_{i-\frac{1}{2}})}{h} = G'(\zeta)$$

for some

$$\zeta \in (x_{i-1}, x_{i+1}).$$

Therefore,

$$\begin{aligned} & \frac{a_{i+\frac{1}{2},j}(u_{i+1,j} - u_{i,j}) - a_{i-\frac{1}{2},j}(u_{i,j} - u_{i-1,j})}{h^2} - F'(x_i) \\ &= \frac{h^2}{24} F'''(\eta) + \frac{h^2}{24} G'(\zeta) + O(h^3). \end{aligned}$$

Taking absolute values gives

$$\begin{aligned} & \left| \frac{a_{i+\frac{1}{2},j}(u_{i+1,j} - u_{i,j}) - a_{i-\frac{1}{2},j}(u_{i,j} - u_{i-1,j})}{h^2} - F'(x_i) \right| \\ & \leq \frac{h^2}{24} \max_{x \in [x_{i-1}, x_{i+1}]} |F'''(x)| + \frac{h^2}{24} \max_{x \in [x_{i-1}, x_{i+1}]} |G'(x)| + O(h^3). \end{aligned}$$

Since

$$F(x) = a(x, y_j) u_x(x, y_j) \in C^3, \quad G(x) = a(x, y_j) u_{xxx}(x, y_j) \in C^1,$$

we have

$$F'''(x) = \frac{\partial^3}{\partial x^3} \left(a \frac{\partial u}{\partial x} \right) (x, y_j),$$

and

$$G'(x) = \frac{\partial}{\partial x} \left(a \frac{\partial^3 u}{\partial x^3} \right) (x, y_j).$$

Thus

$$\begin{aligned} & \left| \frac{a_{i+\frac{1}{2},j}(u_{i+1,j} - u_{i,j}) - a_{i-\frac{1}{2},j}(u_{i,j} - u_{i-1,j})}{h_x^2} - \frac{\partial}{\partial x} \left(a \frac{\partial u}{\partial x} \right) (x_i, y_j) \right| \\ & \leq \frac{h_x^2}{24} \left[\max_{x \in [x_{i-1}, x_{i+1}]} \left| \frac{\partial^3}{\partial x^3} \left(a \frac{\partial u}{\partial x} \right) \right| + \max_{x \in [x_{i-1}, x_{i+1}]} \left| \frac{\partial}{\partial x} \left(a \frac{\partial^3 u}{\partial x^3} \right) \right| \right] + O(h_x^3). \end{aligned}$$

Hence the approximation is second-order accurate. □

PROBLEM II

Let $u(x, y)$ be the solution of Poisson's equation

$$-\Delta u = f$$

on the unit square with homogeneous Dirichlet boundary conditions

$$u(x, 0) = u(x, 1) = u(0, y) = u(1, y) = 0,$$

and let $\mathbf{u}$ be the vector of values $u(x_i, y_j)$ (using natural ordering) on a uniform grid of spacing h in each direction.

Let $\hat{\mathbf{u}}$ be the solution of the linear system

$$-\Delta_h \hat{\mathbf{u}} = \mathbf{f},$$

where $-\Delta_h = A$ is the block tridiagonal matrix

$$A = \begin{pmatrix} S & T & & \\ T & S & T & \\ & \ddots & \ddots & \ddots \\ & & T & S & T \\ & & & T & S \end{pmatrix},$$

with

$$T = -\frac{1}{h^2}I,$$

and

$$S = \begin{pmatrix} d & b & & \\ b & d & b & \\ & \ddots & \ddots & \ddots \\ & & b & d & b \\ & & & b & d \end{pmatrix}, \quad d = \frac{4}{h^2}, \quad b = -\frac{1}{h^2}.$$

Here $\mathbf{f}$ is the vector of right-hand-side values $f(x_i, y_j)$.

Use the previous exercise and the estimate for the eigenvalues of A to show that

$$\sqrt{\frac{1}{n} \sum_{i=1}^n (u_i - \hat{u}_i)^2} \leq Ch^2,$$

for some constant C independent of h .

SOLUTION. Let

$$e = \mathbf{u} - \hat{\mathbf{u}}$$

denote the discretization error.

Since the five-point finite difference approximation of the Laplacian is second-order accurate, the exact solution satisfies

$$-\Delta_h \mathbf{u} = \mathbf{f} + \boldsymbol{\tau},$$

where $\boldsymbol{\tau}$ is the local truncation error vector. Moreover, if $u \in C^4(\Omega)$, then

$$|\tau_i| \leq C_1 h^2, \quad i = 1, \dots, n,$$

for some constant C_1 independent of h . Consequently,

$$\|\boldsymbol{\tau}\|_2 \leq \sqrt{n} C_1 h^2,$$

or equivalently,

$$\frac{1}{\sqrt{n}} \|\boldsymbol{\tau}\|_2 \leq C_1 h^2.$$

Since $A = -\Delta_h$, the sampled exact solution and the numerical solution satisfy

$$A\mathbf{u} = \mathbf{f} + \boldsymbol{\tau}, \quad A\hat{\mathbf{u}} = \mathbf{f}.$$

Subtracting the two equations yields

$$Ae = A(\mathbf{u} - \hat{\mathbf{u}}) = \boldsymbol{\tau}.$$

Since A is nonsingular,

$$e = A^{-1}\boldsymbol{\tau}.$$

Taking the Euclidean norm gives

$$\|e\|_2 \leq \|A^{-1}\|_2 \|\boldsymbol{\tau}\|_2.$$

Therefore, it remains to estimate $\|A^{-1}\|_2$.

For the two-dimensional discrete Laplacian, the eigenvalues are

$$\lambda_{pq} = \frac{4}{h^2} \left(\sin^2 \frac{p\pi}{2(m+1)} + \sin^2 \frac{q\pi}{2(m+1)} \right), \quad 1 \leq p, q \leq m.$$

Hence the smallest eigenvalue is

$$\lambda_{\min}(A) = \frac{8}{h^2} \sin^2 \frac{\pi}{2(m+1)}.$$

Since

$$h = \frac{1}{m+1},$$

we obtain

$$\lambda_{\min}(A) = \frac{8}{h^2} \sin^2 \left(\frac{\pi h}{2} \right).$$

Using the estimate

$$\sin x \geq \frac{2}{\pi} x, \quad 0 \leq x \leq \frac{\pi}{2},$$

with $x = \pi h/2$, we find

$$\lambda_{\min}(A) \geq \frac{8}{h^2} \left(\frac{2\pi h}{\pi} \frac{1}{2} \right)^2 = 8.$$

Therefore

$$\|A^{-1}\|_2 = \frac{1}{\lambda_{\min}(A)} \leq \frac{1}{8},$$

which is bounded independently of h .

Combining this estimate with the truncation-error bound gives

$$\|e\|_2 \leq C \|\tau\|_2.$$

Dividing by $\sqrt{n}$, we obtain

$$\frac{1}{\sqrt{n}} \|e\|_2 \leq C \frac{1}{\sqrt{n}} \|\tau\|_2 \leq Ch^2.$$

Finally,

$$\frac{1}{\sqrt{n}} \|e\|_2 = \sqrt{\frac{1}{n} \sum_{i=1}^n (u_i - \hat{u}_i)^2},$$

and therefore

$$\boxed{\sqrt{\frac{1}{n} \sum_{i=1}^n (u_i - \hat{u}_i)^2} \leq Ch^2.}$$

Thus the discrete solution converges to the exact solution with second-order accuracy in the root-mean-square norm.

□

PROBLEM III For the block-TST matrix A in (1), we know that the eigenvalues are

$$\lambda_{j,k} = \frac{4}{h^2} \sin^2 \left(\frac{j\pi}{2(n+1)} \right) + \frac{4}{h^2} \sin^2 \left(\frac{k\pi}{2(n+1)} \right), \quad j, k = 1, \dots, n,$$

and the corresponding eigenvectors are

$$u_{m,l}^{(j,k)} = \frac{2}{n+1} \sin \left(\frac{mj\pi}{n+1} \right) \sin \left(\frac{lk\pi}{n+1} \right), \quad m, l = 1, \dots, n,$$

where $u_{m,l}^{(j,k)}$ denotes the component corresponding to grid point (m, l) in the eigenvector associated with $\lambda_{j,k}$.

Show that the eigenvectors are orthonormal.

Lemma 2 (Discrete sine orthogonality). For $j, p = 1, \dots, n$,

$$\sum_{m=1}^n \sin\left(\frac{mj\pi}{n+1}\right) \sin\left(\frac{mp\pi}{n+1}\right) = \frac{n+1}{2} \delta_{jp}.$$

证明. Let

$$\theta = \frac{\pi}{n+1}.$$

Using

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i},$$

we have

$$\sin(mj\theta) \sin(mp\theta) = -\frac{1}{4} (e^{im(j+p)\theta} - e^{im(j-p)\theta} - e^{-im(j-p)\theta} + e^{-im(j+p)\theta}).$$

Therefore

$$\sum_{m=1}^n \sin(mj\theta) \sin(mp\theta) = \frac{1}{2} \sum_{m=1}^n \cos(m(j-p)\theta) - \frac{1}{2} \sum_{m=1}^n \cos(m(j+p)\theta).$$

We use the standard finite cosine sum

$$\sum_{m=1}^n \cos(mx) = \frac{\sin\left(\frac{nx}{2}\right) \cos\left(\frac{(n+1)x}{2}\right)}{\sin\left(\frac{x}{2}\right)}.$$

If $j \neq p$, then

$$\sum_{m=1}^n \cos(m(j-p)\theta) = \sum_{m=1}^n \cos(m(j+p)\theta),$$

so

$$\sum_{m=1}^n \sin(mj\theta) \sin(mp\theta) = 0.$$

If $j = p$, then

$$\sum_{m=1}^n \sin^2(mj\theta) = \frac{1}{2} \sum_{m=1}^n (1 - \cos(2mj\theta)).$$

Thus

$$\sum_{m=1}^n \sin^2(mj\theta) = \frac{n}{2} - \frac{1}{2} \sum_{m=1}^n \cos(2mj\theta).$$

Since

$$\sum_{m=1}^n \cos(2mj\theta) = -1,$$

we get

$$\sum_{m=1}^n \sin^2(mj\theta) = \frac{n}{2} + \frac{1}{2} = \frac{n+1}{2}.$$

Hence

$$\sum_{m=1}^n \sin\left(\frac{mj\pi}{n+1}\right) \sin\left(\frac{mp\pi}{n+1}\right) = \frac{n+1}{2} \delta_{jp}.$$

□

SOLUTION. For two eigenvectors corresponding to indices (j, k) and (p, q) , their inner product is

$$\langle u^{(j,k)}, u^{(p,q)} \rangle = \sum_{m=1}^n \sum_{l=1}^n u_{m,l}^{(j,k)} u_{m,l}^{(p,q)}.$$

Substituting the formula of the eigenvectors, we get

$$\begin{aligned} \langle u^{(j,k)}, u^{(p,q)} \rangle &= \sum_{m=1}^n \sum_{l=1}^n \left[\frac{2}{n+1} \sin \left(\frac{mj\pi}{n+1} \right) \sin \left(\frac{lk\pi}{n+1} \right) \right] \\ &\quad \times \left[\frac{2}{n+1} \sin \left(\frac{mp\pi}{n+1} \right) \sin \left(\frac{lq\pi}{n+1} \right) \right] \\ &= \frac{4}{(n+1)^2} \left[\sum_{m=1}^n \sin \left(\frac{mj\pi}{n+1} \right) \sin \left(\frac{mp\pi}{n+1} \right) \right] \\ &\quad \times \left[\sum_{l=1}^n \sin \left(\frac{lk\pi}{n+1} \right) \sin \left(\frac{lq\pi}{n+1} \right) \right]. \end{aligned}$$

Using the discrete sine orthogonality relation

$$\sum_{m=1}^n \sin \left(\frac{mj\pi}{n+1} \right) \sin \left(\frac{mp\pi}{n+1} \right) = \frac{n+1}{2} \delta_{jp},$$

and similarly,

$$\sum_{l=1}^n \sin \left(\frac{lk\pi}{n+1} \right) \sin \left(\frac{lq\pi}{n+1} \right) = \frac{n+1}{2} \delta_{kq},$$

we obtain

$$\begin{aligned} \langle u^{(j,k)}, u^{(p,q)} \rangle &= \frac{4}{(n+1)^2} \cdot \frac{n+1}{2} \delta_{jp} \cdot \frac{n+1}{2} \delta_{kq} \\ &= \delta_{jp} \delta_{kq}. \end{aligned}$$

Therefore,

$$\langle u^{(j,k)}, u^{(p,q)} \rangle = \begin{cases} 1, & (j, k) = (p, q), \\ 0, & (j, k) \neq (p, q). \end{cases}$$

Hence the eigenvectors are orthonormal. □

PROBLEM IV Show that the SSOR preconditioner is

$$M = \frac{\omega}{2 - \omega} (\omega^{-1} D - L) D^{-1} (\omega^{-1} D - U).$$

SOLUTION. Let

$$A = D - L - U,$$

where D is the diagonal part of A , and L, U are respectively the strictly lower and strictly upper triangular parts under the sign convention $A = D - L - U$.

For the forward SOR method, we have the splitting

$$A = M_f - N_f,$$

where

$$M_f = \omega^{-1}D - L, \quad N_f = (\omega^{-1} - 1)D + U.$$

Indeed,

$$M_f - N_f = (\omega^{-1}D - L) - ((\omega^{-1} - 1)D + U) = D - L - U = A.$$

Similarly, for the backward SOR method,

$$A = M_b - N_b,$$

where

$$M_b = \omega^{-1}D - U, \quad N_b = (\omega^{-1} - 1)D + L.$$

The SSOR method consists of one forward SOR sweep followed by one backward SOR sweep. Let $x^{(k+\frac{1}{2})}$ be the intermediate vector after the forward sweep. Then

$$M_f x^{(k+\frac{1}{2})} = N_f x^{(k)} + b,$$

and

$$M_b x^{(k+1)} = N_b x^{(k+\frac{1}{2})} + b.$$

Therefore,

$$x^{(k+\frac{1}{2})} = M_f^{-1} N_f x^{(k)} + M_f^{-1} b,$$

and hence

$$x^{(k+1)} = M_b^{-1} N_b M_f^{-1} N_f x^{(k)} + \dots.$$

Thus the iteration matrix of SSOR is

$$G_{\text{SSOR}} = M_b^{-1} N_b M_f^{-1} N_f.$$

Since

$$N_f = M_f - A, \quad N_b = M_b - A,$$

we get

$$G_{\text{SSOR}} = M_b^{-1} (M_b - A) M_f^{-1} (M_f - A).$$

Expanding gives

$$\begin{aligned} G_{\text{SSOR}} &= (I - M_b^{-1} A)(I - M_f^{-1} A) \\ &= I - M_f^{-1} A - M_b^{-1} A + M_b^{-1} A M_f^{-1} A. \end{aligned}$$

On the other hand, a stationary iteration with splitting

$$A = M_{\text{SSOR}} - N_{\text{SSOR}}$$

has iteration matrix

$$G_{\text{SSOR}} = M_{\text{SSOR}}^{-1} N_{\text{SSOR}} = I - M_{\text{SSOR}}^{-1} A.$$

Comparing this with the expression above, we obtain

$$M_{\text{SSOR}}^{-1} = M_f^{-1} + M_b^{-1} - M_b^{-1} A M_f^{-1}.$$

The right-hand side can be factorized as

$$M_f^{-1} + M_b^{-1} - M_b^{-1} A M_f^{-1} = M_b^{-1} (M_f + M_b - A) M_f^{-1}.$$

Therefore

$$M_{\text{SSOR}}^{-1} = M_b^{-1} (M_f + M_b - A) M_f^{-1}.$$

Taking inverses yields

$$M_{\text{SSOR}} = M_f (M_f + M_b - A)^{-1} M_b.$$

It remains to compute the middle factor. We have

$$\begin{aligned} M_f + M_b - A &= (\omega^{-1} D - L) + (\omega^{-1} D - U) - (D - L - U) \\ &= (2\omega^{-1} - 1) D \\ &= \frac{2 - \omega}{\omega} D. \end{aligned}$$

Hence

$$(M_f + M_b - A)^{-1} = \frac{\omega}{2 - \omega} D^{-1}.$$

Substituting this into the expression for M_{SSOR} , we get

$$\begin{aligned} M_{\text{SSOR}} &= M_f (M_f + M_b - A)^{-1} M_b \\ &= (\omega^{-1} D - L) \left(\frac{\omega}{2 - \omega} D^{-1} \right) (\omega^{-1} D - U) \\ &= \frac{\omega}{2 - \omega} (\omega^{-1} D - L) D^{-1} (\omega^{-1} D - U). \end{aligned}$$

Therefore

$$M_{\text{SSOR}} = \frac{\omega}{2 - \omega} (\omega^{-1} D - L) D^{-1} (\omega^{-1} D - U).$$

This proves the desired formula. $\square$

PROBLEM V Show that if a matrix A has Property A, then there is a permutation matrix P such that $P^{-1} A P$ satisfies

$$\det(cI - D^{-1} L - D^{-1} U) = \det(cI - \gamma D^{-1} L - \gamma^{-1} D^{-1} U) \quad (2)$$

for all c and $\gamma \neq 0$.

SOLUTION. Since A has Property A, there exists a permutation matrix P such that

$$\tilde{A} = P^{-1} A P$$

is consistently ordered. Consequently, the index set can be partitioned into two disjoint subsets, and $\tilde{A}$ takes the block form

$$\tilde{A} = \begin{pmatrix} D_1 & -B \\ -C & D_2 \end{pmatrix},$$

where D_1 and D_2 are diagonal matrices. Indeed, Property A implies that there are no off-diagonal couplings between unknowns belonging to the same subset, so the diagonal blocks contain only diagonal entries.

Let

$$D = \begin{pmatrix} D_1 & 0 \\ 0 & D_2 \end{pmatrix}, \quad L = \begin{pmatrix} 0 & 0 \\ C & 0 \end{pmatrix}, \quad U = \begin{pmatrix} 0 & B \\ 0 & 0 \end{pmatrix}.$$

Then

$$\tilde{A} = D - L - U.$$

Since D_1 and D_2 are diagonal,

$$D^{-1} = \begin{pmatrix} D_1^{-1} & 0 \\ 0 & D_2^{-1} \end{pmatrix},$$

and therefore

$$D^{-1}L = \begin{pmatrix} 0 & 0 \\ D_2^{-1}C & 0 \end{pmatrix}, \quad D^{-1}U = \begin{pmatrix} 0 & D_1^{-1}B \\ 0 & 0 \end{pmatrix}.$$

Now define

$$T = \begin{pmatrix} \gamma^{1/2}I & 0 \\ 0 & \gamma^{-1/2}I \end{pmatrix},$$

which is nonsingular for $\gamma \neq 0$.

A direct computation yields

$$T^{-1}(D^{-1}L)T = \begin{pmatrix} 0 & 0 \\ \gamma D_2^{-1}C & 0 \end{pmatrix} = \gamma D^{-1}L,$$

and similarly

$$T^{-1}(D^{-1}U)T = \begin{pmatrix} 0 & \gamma^{-1}D_1^{-1}B \\ 0 & 0 \end{pmatrix} = \gamma^{-1}D^{-1}U.$$

Hence

$$T^{-1}(cI - D^{-1}L - D^{-1}U)T = cI - \gamma D^{-1}L - \gamma^{-1}D^{-1}U.$$

Thus

$$cI - D^{-1}L - D^{-1}U$$

and

$$cI - \gamma D^{-1}L - \gamma^{-1}D^{-1}U$$

are similar matrices. Taking determinants gives

$$\det(cI - D^{-1}L - D^{-1}U) = \det(cI - \gamma D^{-1}L - \gamma^{-1}D^{-1}U).$$

Therefore $P^{-1}AP$ satisfies the stated identity for all c and $\gamma \neq 0$. □

PROBLEM VI (Verify Young's theory by experiment)

- (a) Show that the 5-point approximation matrix A in the form (1) has Property A.
- (b) Let

$$n = 30, \quad h = \frac{1}{n+1}.$$

Compute the spectral radius (thus the approximate convergence rate) of the Jacobi iteration matrix, the Gauss–Seidel iteration matrix, and the SOR iteration matrix with the optimal value of ω .

Moreover, compute the condition number of A and thus the approximate convergence rate of the CG method.

- (c) Plot to verify the above results and comment.

SECTION. 1. We consider the standard five-point finite difference matrix for the two-dimensional Poisson equation. The unknowns are located at the grid points

$$(i, j), \quad 1 \leq i, j \leq n.$$

For each interior grid point, the five-point stencil couples (i, j) only with its four nearest neighbors

$$(i-1, j), \quad (i+1, j), \quad (i, j-1), \quad (i, j+1).$$

To prove that A has Property A, we use a red–black ordering of the grid points. Define

$$\mathcal{R} = \{(i, j) : i + j \text{ is even}\}, \quad \mathcal{B} = \{(i, j) : i + j \text{ is odd}\}.$$

If two grid points are nearest neighbors, then their indices differ by one in exactly one coordinate. Hence the parity of $i + j$ changes. Therefore, every neighbor of a point in $\mathcal{R}$ lies in $\mathcal{B}$, and every neighbor of a point in $\mathcal{B}$ lies in $\mathcal{R}$.

Thus there is no coupling between two red points and no coupling between two black points. After permuting the unknowns so that all red points come first and all black points come second, the matrix A takes the block form

$$PAP^T = \begin{pmatrix} D_1 & -B \\ -C & D_2 \end{pmatrix},$$

where D_1 and D_2 are diagonal matrices corresponding to the diagonal entries of A , and the off-diagonal blocks contain all couplings between red and black grid points.

Therefore the five-point approximation matrix A has Property A.

2.

$$n = 30, \quad h = \frac{1}{n+1} = \frac{1}{31}.$$

For the five-point approximation matrix, the Jacobi iteration matrix is

$$G_J = D^{-1}(L + U).$$

Its eigenvalues are

$$\mu_{pq} = \frac{1}{2} \left(\cos \frac{p\pi}{n+1} + \cos \frac{q\pi}{n+1} \right), \quad p, q = 1, \dots, n.$$

Hence

$$\rho(G_J) = \cos \frac{\pi}{n+1} = \cos \frac{\pi}{31} \approx 0.994869.$$

Since the five-point matrix has Property A, Young's theorem gives

$$\rho(G_{GS}) = \rho(G_J)^2.$$

Therefore

$$\rho(G_{GS}) = \cos^2 \frac{\pi}{31} \approx 0.989765.$$

For SOR, the optimal relaxation parameter is

$$\omega_{\text{opt}} = \frac{2}{1 + \sqrt{1 - \rho(G_J)^2}}.$$

Thus

$$\omega_{\text{opt}} = \frac{2}{1 + \sqrt{1 - \cos^2(\pi/31)}} = \frac{2}{1 + \sin(\pi/31)} \approx 1.816253.$$

The corresponding optimal SOR spectral radius is

$$\rho(G_{SOR}) = \omega_{\text{opt}} - 1 \approx 0.816253.$$

Next, the eigenvalues of the five-point matrix are

$$\lambda_{pq} = \frac{4}{h^2} \left(\sin^2 \frac{p\pi}{2(n+1)} + \sin^2 \frac{q\pi}{2(n+1)} \right), \quad p, q = 1, \dots, n.$$

Hence

$$\lambda_{\min} = \frac{8}{h^2} \sin^2 \frac{\pi}{2(n+1)},$$

and

$$\lambda_{\max} = \frac{8}{h^2} \cos^2 \frac{\pi}{2(n+1)}.$$

Therefore

$$\kappa(A) = \frac{\lambda_{\max}}{\lambda_{\min}} = \cot^2 \frac{\pi}{2(n+1)}.$$

For $n = 30$,

$$\kappa(A) = \cot^2 \frac{\pi}{62} \approx 388.812134.$$

The CG convergence factor is estimated by

$$q_{CG} = \frac{\sqrt{\kappa(A)} - 1}{\sqrt{\kappa(A)} + 1}.$$

Thus

$$q_{CG} \approx \frac{\sqrt{388.812134} - 1}{\sqrt{388.812134} + 1} \approx 0.903467.$$

In summary,

$$\rho(G_J) \approx 0.994869,$$

$$\rho(G_{GS}) \approx 0.989765,$$

$$\omega_{\text{opt}} \approx 1.816253, \quad \rho(G_{SOR}) \approx 0.816253,$$

and

$$\kappa(A) \approx 388.812134, \quad q_{CG} \approx 0.903467.$$

3. Theoretical convergence factors obtained in part (b) are

$$\rho_J = 0.99486932, \quad \rho_{GS} = 0.98976497, \quad \rho_{SOR} = 0.81625276.$$

The residual histories of Jacobi, Gauss–Seidel, and optimal SOR are shown in Figure 1.

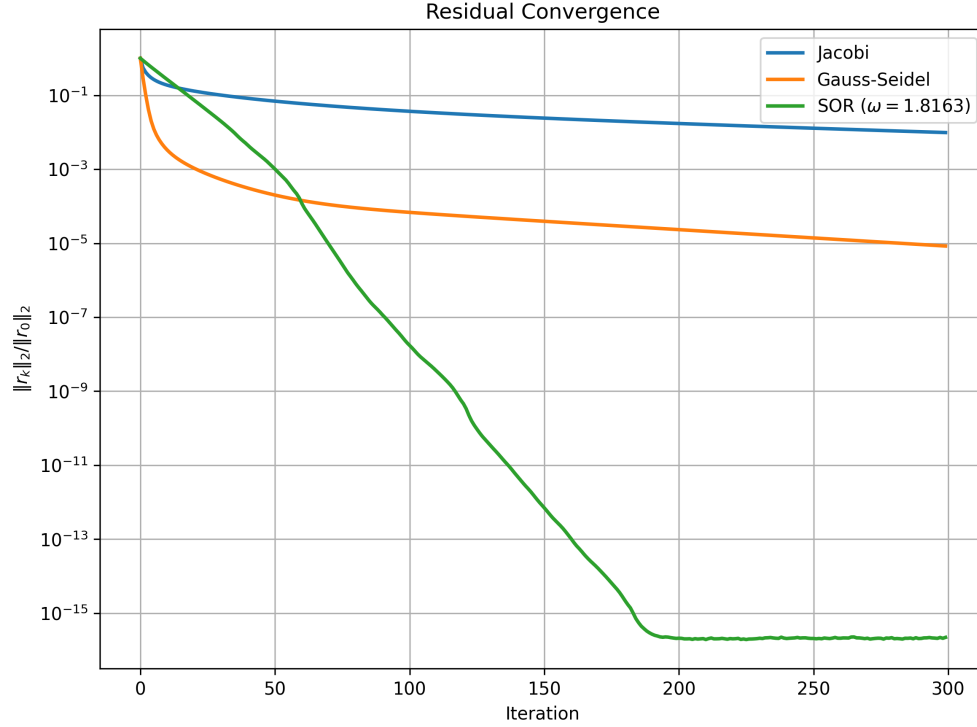


图 1: Relative residual histories of Jacobi, Gauss–Seidel and optimal SOR.

From Figure 1, the Jacobi iteration converges most slowly, the Gauss–Seidel iteration is slightly faster, and the optimal SOR iteration converges significantly faster than both. This agrees with the theoretical ordering

$$\rho_{SOR} < \rho_{GS} < \rho_J.$$

To further verify Young’s theory, we plot

$$q_k = \frac{\|r_{k+1}\|_2}{\|r_k\|_2},$$

together with the theoretical spectral radii.

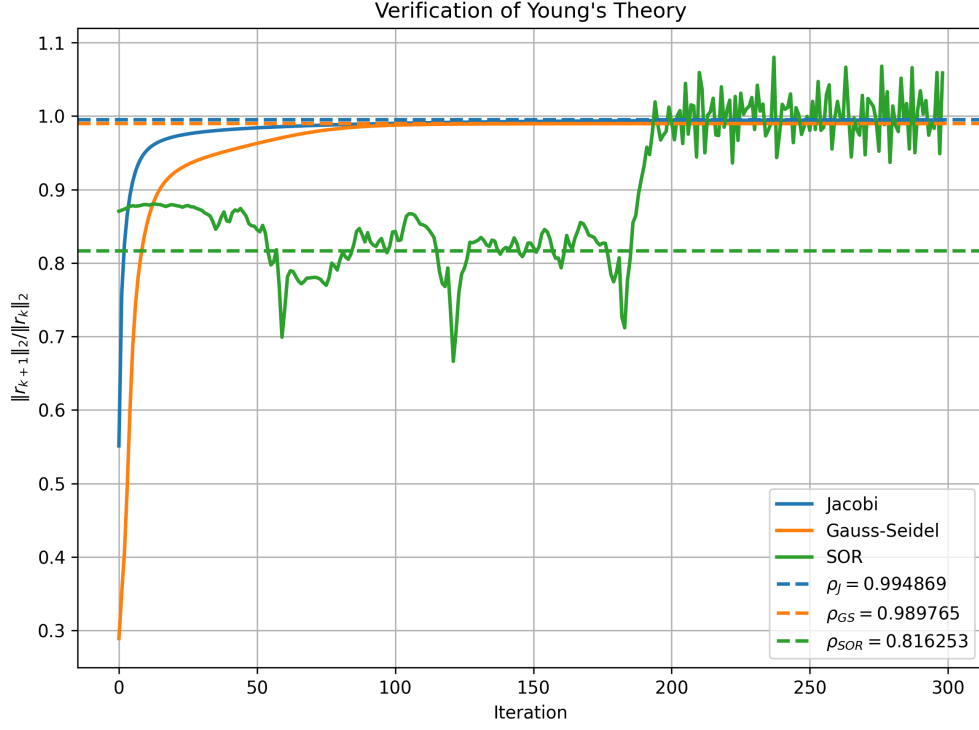


图 2: Observed convergence factors and theoretical spectral radii.

For Jacobi and Gauss–Seidel, the observed asymptotic convergence factors are

$$q_J \approx 0.99459726, \quad q_{GS} \approx 0.98975788,$$

which are very close to

$$\rho_J = 0.99486932, \quad \rho_{GS} = 0.98976497.$$

In particular,

$$\rho_{GS} = \rho_J^2,$$

which is exactly the relation predicted by Young's theorem for matrices having Property A. For the SOR method, the residual decreases rapidly to the level of machine precision. Consequently, the quantity

$$\frac{\|r_{k+1}\|_2}{\|r_k\|_2}$$

is eventually dominated by round-off errors and no longer provides a reliable estimate of the asymptotic convergence factor. Therefore, only the initial asymptotic regime should be used for comparison with the theoretical value

$$\rho_{SOR} = 0.81625276.$$

Overall, the numerical results agree with Young's theory and confirm the predicted convergence behavior of Jacobi, Gauss–Seidel, and optimal SOR iterations for the five-point approximation matrix.

□

PROBLEM VII

$$1. |A^k| \leq |A|^k \quad \text{for all } k = 1, 2, \dots$$

$$2. \text{ If } 0 \leq A \leq B, \text{ then}$$

$$0 \leq A^k \leq B^k \quad \text{for all } k = 1, 2, \dots$$

$$3. \text{ If } A > 0, \text{ then}$$

$$A^k > 0 \quad \text{for all } k = 1, 2, \dots$$

$$4. \text{ If } A > 0, v \geq 0, \text{ and } v \text{ is not the zero vector, then } Av > 0.$$

$$5. \text{ If } A \geq 0, v \geq 0, \text{ and } Av \geq \alpha v \text{ for some } \alpha > 0, \text{ then}$$

$$A^k v \geq \alpha^k v \quad \text{for all } k = 1, 2, \dots$$

SOLUTION. Let $A = (a_{ij})$.

(a) We prove that

$$|A^k| \leq |A|^k, \quad k = 1, 2, \dots$$

For $k = 1$, the result is obvious. Suppose that

$$|A^k| \leq |A|^k.$$

Then

$$|A^{k+1}| = |A^k A| \leq |A^k| |A| \leq |A|^k |A| = |A|^{k+1}.$$

Hence, by induction,

$$|A^k| \leq |A|^k$$

for all $k = 1, 2, \dots$

(b) Assume that

$$0 \leq A \leq B.$$

We prove that

$$0 \leq A^k \leq B^k, \quad k = 1, 2, \dots$$

Since $A \geq 0$, every entry of A^k is a sum of products of nonnegative numbers. Hence

$$A^k \geq 0.$$

Now for each i, j , we write the (i, j) -entry of A^k as

$$(A^k)_{ij} = \sum_{i_1, \dots, i_{k-1}} a_{ii_1} a_{i_1 i_2} \cdots a_{i_{k-1} j}.$$

Similarly,

$$(B^k)_{ij} = \sum_{i_1, \dots, i_{k-1}} b_{ii_1} b_{i_1 i_2} \cdots b_{i_{k-1} j}.$$

Because

$$0 \leq a_{pq} \leq b_{pq}$$

for all p, q , we have

$$a_{ii_1} a_{i_1 i_2} \cdots a_{i_{k-1} j} \leq b_{ii_1} b_{i_1 i_2} \cdots b_{i_{k-1} j}.$$

Therefore,

$$(A^k)_{ij} \leq (B^k)_{ij}.$$

Since this holds for every i, j , we obtain

$$0 \leq A^k \leq B^k.$$

Thus

$$0 \leq A^k \leq B^k, \quad k = 1, 2, \dots$$

(c) Assume

$$A > 0.$$

We prove that

$$A^k > 0, \quad k = 1, 2, \dots$$

For $k = 1$, this is the assumption. Suppose $A^k > 0$. Then for every i, j ,

$$(A^{k+1})_{ij} = (A^k A)_{ij} = \sum_{\ell} (A^k)_{i\ell} a_{\ell j}.$$

Since $(A^k)_{i\ell} > 0$ and $a_{\ell j} > 0$, every term in the sum is positive. Hence

$$(A^{k+1})_{ij} > 0.$$

Thus $A^{k+1} > 0$. By induction,

$$A^k > 0$$

for all $k = 1, 2, \dots$

(d) Assume

$$A > 0, \quad v \geq 0, \quad v \neq 0.$$

We prove that

$$Av > 0.$$

Since $v \geq 0$ and $v \neq 0$, there exists an index j_0 such that

$$v_{j_0} > 0.$$

For each i ,

$$(Av)_i = \sum_j a_{ij}v_j.$$

Because $A > 0$, we have $a_{ij_0} > 0$. Therefore

$$(Av)_i = \sum_j a_{ij}v_j \geq a_{ij_0}v_{j_0} > 0.$$

Hence every component of Av is positive, so

$$Av > 0.$$

(e) Assume that

$$A \geq 0, \quad v \geq 0, \quad Av \geq \alpha v,$$

where $\alpha > 0$. We prove that

$$A^k v \geq \alpha^k v, \quad k = 1, 2, \dots$$

For $k = 1$, the conclusion is exactly the assumption:

$$Av \geq \alpha v.$$

Now suppose that for some $k \geq 1$,

$$A^k v \geq \alpha^k v.$$

Then

$$A^k v - \alpha^k v \geq 0.$$

Since $A \geq 0$, multiplying a nonnegative vector by A still gives a nonnegative vector. Hence

$$A(A^k v - \alpha^k v) \geq 0.$$

Therefore

$$A^{k+1} v - \alpha^k Av \geq 0,$$

so

$$A^{k+1} v \geq \alpha^k Av.$$

Using the assumption $Av \geq \alpha v$, we get

$$A^{k+1} v \geq \alpha^k Av \geq \alpha^k \alpha v = \alpha^{k+1} v.$$

Thus, by induction,

$$A^k v \geq \alpha^k v, \quad k = 1, 2, \dots$$

□

PROBLEM VIII Show that the iterates x_k generated by the PCG algorithm with preconditioner M are the same as those generated with preconditioner cM for any $c > 0$.

We provide 2 perspective to solve the problem.

SOLUTION. Consider the PCG method applied to the linear system

$$Ax = b,$$

with preconditioner M . Let

$$r_k = b - Ax_k, \quad Mz_k = r_k.$$

Equivalently,

$$z_k = M^{-1}r_k.$$

If the preconditioner is changed from M to cM , where $c > 0$, then the corresponding preconditioned residual $\tilde{z}_k$ satisfies

$$(cM)\tilde{z}_k = r_k.$$

Hence

$$\tilde{z}_k = (cM)^{-1}r_k = \frac{1}{c}M^{-1}r_k = \frac{1}{c}z_k.$$

In the PCG algorithm, the step size is

$$\alpha_k = \frac{r_k^T z_k}{p_k^T A p_k}.$$

For the preconditioner cM , the numerator becomes

$$r_k^T \tilde{z}_k = r_k^T \left(\frac{1}{c} z_k \right) = \frac{1}{c} r_k^T z_k.$$

We now show that the search directions are scaled by the same factor. For $k = 0$,

$$\tilde{p}_0 = \tilde{z}_0 = \frac{1}{c} z_0 = \frac{1}{c} p_0.$$

Assume that

$$\tilde{p}_k = \frac{1}{c} p_k.$$

Then

$$\tilde{p}_k^T A \tilde{p}_k = \left(\frac{1}{c} p_k \right)^T A \left(\frac{1}{c} p_k \right) = \frac{1}{c^2} p_k^T A p_k.$$

Therefore the step size for the preconditioner cM is

$$\tilde{\alpha}_k = \frac{r_k^T \tilde{z}_k}{\tilde{p}_k^T A \tilde{p}_k} = \frac{(1/c) r_k^T z_k}{(1/c^2) p_k^T A p_k} = c \alpha_k.$$

Thus

$$\tilde{\alpha}_k \tilde{p}_k = (c \alpha_k) \left(\frac{1}{c} p_k \right) = \alpha_k p_k.$$

Hence

$$\tilde{x}_{k+1} = \tilde{x}_k + \tilde{\alpha}_k \tilde{p}_k = x_k + \alpha_k p_k = x_{k+1}.$$

So the iterates x_k are unchanged.

It remains to check that the next search direction has the same scaling. The residual update is

$$r_{k+1} = r_k - \alpha_k A p_k.$$

Since

$$\tilde{\alpha}_k \tilde{p}_k = \alpha_k p_k,$$

we also have

$$\tilde{r}_{k+1} = r_{k+1}.$$

Therefore

$$\tilde{z}_{k+1} = (cM)^{-1} r_{k+1} = \frac{1}{c} z_{k+1}.$$

The coefficient β_k is

$$\beta_k = \frac{r_{k+1}^T z_{k+1}}{r_k^T z_k}.$$

For the scaled preconditioner,

$$\tilde{\beta}_k = \frac{r_{k+1}^T \tilde{z}_{k+1}}{r_k^T \tilde{z}_k} = \frac{(1/c) r_{k+1}^T z_{k+1}}{(1/c) r_k^T z_k} = \beta_k.$$

Thus

$$\tilde{p}_{k+1} = \tilde{z}_{k+1} + \tilde{\beta}_k \tilde{p}_k = \frac{1}{c} z_{k+1} + \beta_k \frac{1}{c} p_k = \frac{1}{c} p_{k+1}.$$

By induction, the search directions are scaled by $1/c$, while the step sizes are scaled by c . Their product is unchanged, so the iterates x_k generated by PCG with preconditioner M and cM are the same. $\square$

SOLUTION. Let

$$r_0 = b - Ax_0.$$

For the preconditioner M , the preconditioned matrix is

$$B = M^{-1}A.$$

For the scaled preconditioner cM , $c > 0$, the corresponding preconditioned matrix is

$$\tilde{B} = (cM)^{-1}A = \frac{1}{c} M^{-1}A = \frac{1}{c} B.$$

The k -th Krylov subspace generated by B is

$$\mathcal{K}_k(B, r_0) = \text{span}\{r_0, Br_0, \dots, B^{k-1}r_0\}.$$

Similarly,

$$\mathcal{K}_k(\tilde{B}, r_0) = \text{span}\{r_0, \tilde{B}r_0, \dots, \tilde{B}^{k-1}r_0\}.$$

Since

$$\tilde{B}^j = \left(\frac{1}{c}B\right)^j = \frac{1}{c^j}B^j,$$

we obtain

$$\mathcal{K}_k(\tilde{B}, r_0) = \text{span} \left\{ r_0, \frac{1}{c}Br_0, \dots, \frac{1}{c^{k-1}}B^{k-1}r_0 \right\}.$$

Multiplying basis vectors by nonzero scalars does not change the span, hence

$$\mathcal{K}_k(\tilde{B}, r_0) = \mathcal{K}_k(B, r_0).$$

Therefore both PCG methods generate exactly the same Krylov subspaces.

A fundamental characterization of PCG states that x_k is the unique vector in

$$x_0 + \mathcal{K}_k(B, r_0)$$

whose error is A -orthogonal to the Krylov subspace; equivalently,

$$x_k = \arg \min_{x \in x_0 + \mathcal{K}_k(B, r_0)} \|x - x_*\|_A.$$

Since the two methods have the same initial guess x_0 , the same matrix A , and the same Krylov subspace at every iteration, the minimization problem defining x_k is identical in both cases.

Consequently,

$$\tilde{x}_k = x_k, \quad k = 0, 1, 2, \dots$$

Hence the iterates generated by PCG with preconditioners M and cM are identical. $\square$

PROBLEM IX The multigroup transport equation can be written in the form

$$\Omega \cdot \nabla \psi_g + \sigma_g \psi_g - \sum_{g'=1}^G \int_{S^2} \sigma_{g,g'}(r, \Omega \cdot \Omega') \psi_{g'}(r, \Omega') d\Omega' = f_g, \quad g = 1, \dots, G,$$

where $\psi_g(r, \Omega)$ is the unknown flux associated with energy group g and $\sigma_g(r, \Omega)$, $\sigma_{g,g'}(r, \Omega \cdot \Omega')$, and $f_g(r, \Omega)$ are known cross-section and source terms. (Appropriate boundary conditions are also given.)

A standard method for solving this set of equations is to move the terms of the sum corresponding to different energy groups to the right-hand side and solve the resulting set of equations for $\psi_1, \dots, \psi_G$ in increasing order of index, using the most recently updated quantities on the right-hand side; that is,

$$(\Omega \cdot \nabla + \sigma_g) \psi_g^{(k)} - \int_{S^2} \sigma_{g,g} \psi_g^{(k)} d\Omega' = f_g + \sum_{g' < g} \int_{S^2} \sigma_{g,g'} \psi_{g'}^{(k)} d\Omega' + \sum_{g' > g} \int_{S^2} \sigma_{g,g'} \psi_{g'}^{(k-1)} d\Omega'.$$

Identify this procedure with one of the preconditioned iterative methods described in the lectures. How might it be accelerated?

SOLUTION. Let

$$\psi = (\psi_1, \psi_2, \dots, \psi_G)^T$$

be the vector of all unknown group fluxes. This means that instead of treating the G equations separately, we regard all fluxes as one large unknown vector.

Define the diagonal block operator

$$A_{gg}\psi_g = (\Omega \cdot \nabla + \sigma_g)\psi_g - \int_{S^2} \sigma_{g,g}\psi_g d\Omega'.$$

This is the part of the g -th equation that involves only the unknown ψ_g itself, so it represents the within-group transport operator.

For $g' \neq g$, define the off-diagonal coupling operators

$$A_{gg'}\psi_{g'} = - \int_{S^2} \sigma_{g,g'}\psi_{g'} d\Omega'.$$

These terms describe the coupling from energy group g' to energy group g .

With these definitions, the multigroup transport equations can be written as one block linear system

$$A\psi = f.$$

Now split the block operator A as

$$A = D - L - U,$$

where D contains the diagonal blocks A_{gg} , L contains the strictly lower-triangular block part, and U contains the strictly upper-triangular block part.

In the given procedure, when solving for $\psi_g^{(k)}$, the quantities $\psi_{g'}^{(k)}$ with $g' < g$ have already been updated in the current sweep, while the quantities $\psi_{g'}^{(k-1)}$ with $g' > g$ have not yet been updated and therefore are taken from the previous iteration.

Therefore the iteration can be written in block form as

$$(D - L)\psi^{(k)} = f + U\psi^{(k-1)}.$$

Equivalently,

$$\psi^{(k)} = (D - L)^{-1}f + (D - L)^{-1}U\psi^{(k-1)}.$$

This is exactly the block Gauss-Seidel iteration, because the unknowns are updated successively in the order $g = 1, 2, \dots, G$, and the newest available values are immediately used.

From the viewpoint of preconditioned iterative methods, set

$$M = D - L.$$

Then the iteration can also be written as

$$\psi^{(k)} = \psi^{(k-1)} + M^{-1}(f - A\psi^{(k-1)}).$$

Indeed, since

$$A = D - L - U = M - U,$$

we have

$$f - A\psi^{(k-1)} = f - (M - U)\psi^{(k-1)} = f - M\psi^{(k-1)} + U\psi^{(k-1)}.$$

Hence

$$\psi^{(k-1)} + M^{-1}(f - A\psi^{(k-1)}) = M^{-1}f + M^{-1}U\psi^{(k-1)},$$

which is the same iteration as above.

Thus the procedure is a preconditioned Richardson iteration with the block Gauss–Seidel preconditioner

$$M = D - L.$$

Equivalently, it is the block Gauss–Seidel method applied to the multigroup transport system.

A natural way to accelerate the procedure is to use the same operator

$$M = D - L$$

as a preconditioner in a Krylov subspace method, such as GMRES, GCR, or BiCGSTAB. For example, one may solve the left-preconditioned system

$$M^{-1}A\psi = M^{-1}f.$$

Applying M^{-1} corresponds to one forward sweep through the energy groups, so the preconditioner is natural and relatively inexpensive. The Krylov method then accelerates the convergence compared with the stationary block Gauss–Seidel iteration.

Therefore,

the procedure is block Gauss–Seidel, or equivalently preconditioned Richardson with $M = D - L$,

and it can be accelerated by Krylov methods such as GMRES using this block Gauss–Seidel sweep as a preconditioner. $\square$